1. $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n (1)z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \qquad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

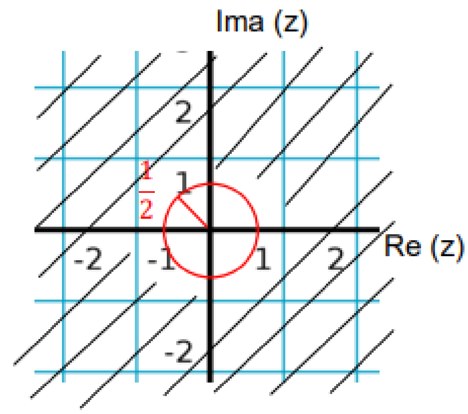
$$\boxed{x[z] = \frac{1}{1-\frac{1}{2}z^{-1}}}$$

ROC:

$$\left|\frac{1}{2}z^{-1}\right| < 1$$

$$\left|\frac{1}{2z}\right| < 1$$

$$\boxed{|z| > \frac{1}{2}}$$



2. $x[n] = u[n]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=0}^{\infty} (1)z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n, \qquad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

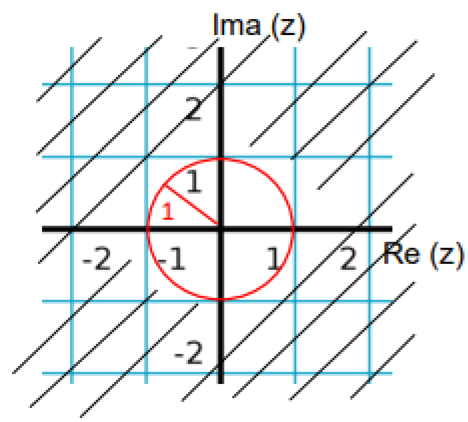
$$\boxed{x[z] = \frac{1}{1-z^{-1}}}$$

ROC:

$$|z^{-1}| < 1$$

$$\left|\frac{1}{z}\right| < 1$$

$$\boxed{|z| > 1}$$



3. $x[n] = -u[-n-1]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} (-1)z^{-n}$$

$$= -\sum_{n=1}^{\infty} (z)^{n}, \qquad n = -n$$

$$= -\sum_{m=0}^{\infty} (z)^{m+1}, \qquad n-1 = m$$
$$n = m+1$$

$$= -\sum_{m=0}^{\infty} z^{m} z$$

$$= -z\sum_{m=0}^{\infty} z^{m}, \qquad \sum_{n=0}^{\infty} x^{n} = \frac{1}{1-x}$$

$$= \frac{-z}{1-z} = \left[\frac{-z}{1-z}\right]\frac{-z}{-z}$$

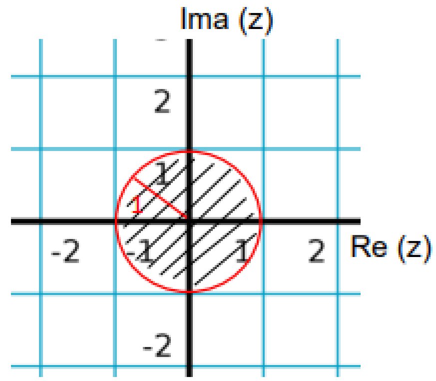
$$\boxed{x[z] = \frac{1}{1-\frac{1}{z}}}$$

ROC:

$|z^{-1}| < 1$

$\left|\frac{1}{z}\right| < 1$

$\boxed{|z| < 1}$



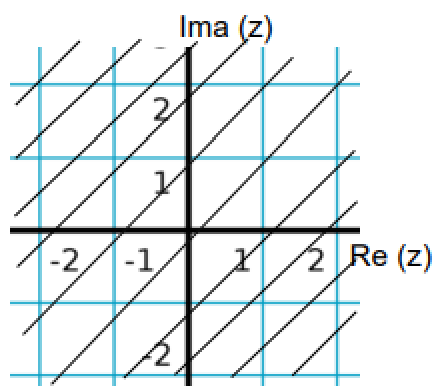
4. $x[n] = \delta[n]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \delta[n]z^{-n}$$

$$= z^{-n}|n = 0$$

$$\boxed{x[z] = z^{0} = 1}$$



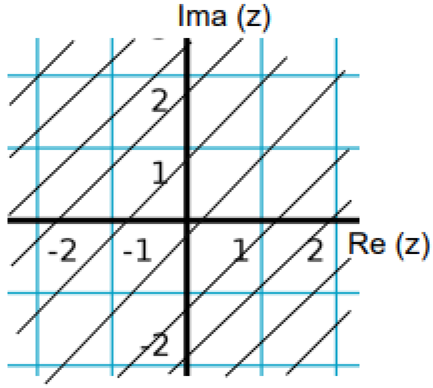
5. $x[n] = \delta[n-k]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \delta[n] z^{-n}$$

$$= z^{-n} | n = k$$

$$\boxed{x[z] = z^{-k}}$$



z = 0 when k > 0

z = ∞ when k < 0

6. $x[n] = 7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n u[n]$

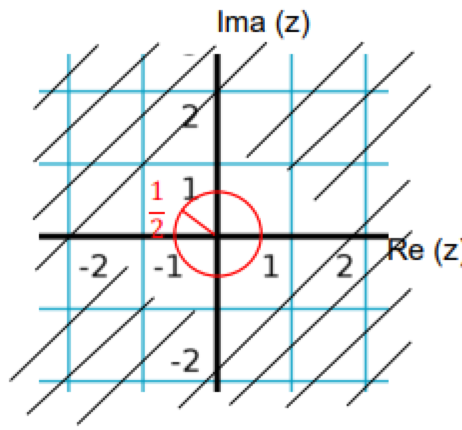
$$x[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} \left[7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n\right] z^{-n}$$

$$= 7\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} - 6\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}$$

$$= 7\sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n - 6\sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$\boxed{x[z] = \frac{7}{1-\frac{1}{3}z^{-1}} - \frac{6}{1-\frac{1}{2}z^{-1}}}$$



ROC₁:

$$\left|\frac{1}{3} z^{-1}\right| < 1$$

$$\left|\frac{1}{3z}\right| < 1$$

$$\boxed{|z| > \frac{1}{3}}$$

ROC₂:

$$\left|\frac{1}{2} z^{-1}\right| < 1$$

$$\left|\frac{1}{2z}\right| < 1$$

$$\boxed{|z| > \frac{1}{2}}$$

7. $x[n] = -u[-n-1] + \left(\frac{1}{2}\right)^n u[n]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \left[-u[-n-1] + \left(\frac{1}{2}\right)^n u[n]\right] z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} z^{-n} + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}, \qquad n = -n$$

$$= -\sum_{m=0}^{\infty} z^{n} + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \qquad \begin{array}{c} n-1=m \\ n=m+1 \end{array}$$

$$= -\sum_{m=0}^{\infty} z^{m+1} + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n$$

$$= -\sum_{m=0}^{\infty} z^{m} z + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n$$

$$= -z\sum_{m=0}^{\infty} z^{m} + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \qquad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$= \frac{-z}{1-z} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{-z}{-z}\left[\frac{-z}{1-z}\right] + \frac{1}{1-\frac{1}{2}z^{-1}}$$

$$\boxed{x[z] = \frac{1}{1-\frac{1}{z}} + \frac{1}{1-\frac{1}{2}z^{-1}}}$$

ROC$_1$:

$|z^{-1}| < 1$

$\left|\frac{1}{z}\right| < 1$

$|z| < 1$

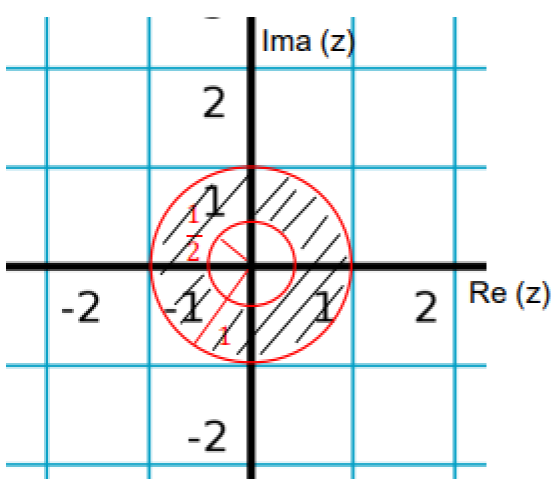
ROC$_2$:

$\left|\frac{1}{2}z^{-1}\right| < 1$

$\left|\frac{1}{2z}\right| < 1$

$|z| > \frac{1}{2}$

$$\boxed{\text{ROC: } \frac{1}{2} < |z| < 1}$$



$$x[z] = \left[\frac{1}{1-\frac{1}{z}} + \frac{1}{1-\frac{1}{2}z^{-1}}\right]\frac{z}{z}$$

$$= \frac{z}{z-1} + \frac{z}{z-\frac{1}{2}}$$

$$= \frac{z\left(z-\frac{1}{2}\right) + z(z-1)}{(z-1)\left(z-\frac{1}{2}\right)}$$

$$= \frac{z^2 - \frac{1}{2}z + z^2 - z}{(z-1)\left(z-\frac{1}{2}\right)}$$

$$= \frac{2z^2 - \frac{3}{2}z}{(z-1)\left(z-\frac{1}{2}\right)}$$

$$= \frac{z\left(2z-\frac{3}{2}\right)}{(z-1)\left(z-\frac{1}{2}\right)}$$

Zeroes

$$\boxed{z = 0}$$

$$2z - \frac{3}{2} = 0$$

$$\boxed{z = \frac{3}{4}}$$

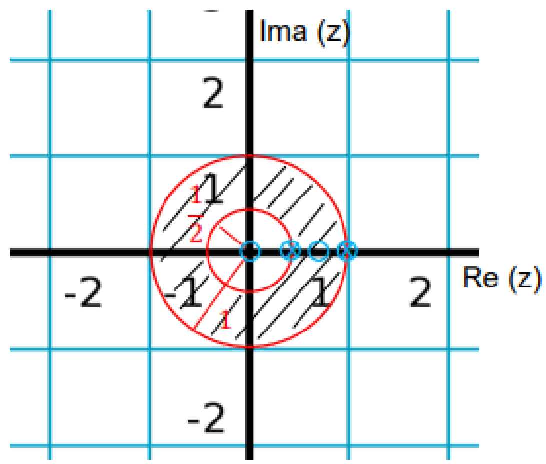
Poles

$$z - 1 = 0$$

$$\boxed{z = 1}$$

$$z - \frac{1}{2} = 0$$

$$\boxed{z = \frac{1}{2}}$$



8. $x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n]$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} \left[\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n]\right] z^{-n}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n \left(\frac{e^{\frac{j\pi n}{4}} - e^{\frac{-j\pi n}{4}}}{2j}\right) z^{-n}$$

$$= \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}\right)^n - \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{\frac{-j\pi}{4}} z^{-1}\right)^n$$

$$\boxed{x[z] = \frac{1}{2j}\left(\frac{1}{1 - \frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}}\right) - \frac{1}{2j}\left(\frac{1}{1 - \frac{1}{3} e^{\frac{-j\pi}{4}} z^{-1}}\right)}$$

ROC$_1$:

$$\left|\frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}\right| < 1$$

$$e^{\pm\theta} = \cos\theta \pm j \sin\theta$$

$$e^{\frac{j\pi}{4}} = \cos\frac{j\pi}{4} + j \sin\frac{j\pi}{4}$$

$$= \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}$$

$$\left|e^{\frac{j\pi}{4}}\right| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = 1$$

$$\left|\frac{1}{3z}\right| < 1$$

$$\boxed{|z| > \frac{1}{3}}$$

ROC$_2$:

$$\left|\frac{1}{3} e^{\frac{-j\pi}{4}} z^{-1}\right| < 1$$

$$e^{\pm\theta} = \cos\theta \pm j \sin\theta$$

$$\left|\frac{1}{3z}\right| < 1$$

$$\boxed{|z| > \frac{1}{3}}$$